BASIC PROBABILITY FOR QUANTUM COMPUTING

Events, conditional probability, random variables, distributions, expectation, measurement statistics, and finite-shot estimation

A detailed reference connecting classical probability theory to the Born rule, quantum measurements, correlations, noise, and experimental data.

Topic 7 in the Quantum Mathematics Foundation series

How to use this volume

Probability is the language used to connect a quantum state to experimental outcomes. The mathematics is classical probability theory; what is specifically quantum is the rule that generates the probabilities and the way amplitudes interfere before those probabilities are formed.

Prerequisites: complex numbers, vectors, inner products, basic matrices, bras and kets, and projectors.

Learning objectives

- Define sample spaces, events, conditional probability, independence, and Bayes theorem.
- Work with discrete random variables, probability distributions, expectation, variance, and covariance.
- Analyze Bernoulli, binomial, multinomial, and finite-shot measurement data.
- Use the Born rule to calculate quantum measurement probabilities in different bases.
- Compute expectation values and correlations from theoretical states and experimental counts.
- Distinguish a coherent superposition from a classical probabilistic mixture.
- Estimate sampling uncertainty and understand basic readout-error models.

Contents

1. Probability as a mathematical model

Probability is not simply "uncertainty" in everyday language. It is a mathematical model that assigns numbers to events and obeys strict consistency rules. The same framework can describe dice, communication errors, noisy sensors, randomized algorithms, and repeated quantum measurements.

A complete probability model contains three ingredients:

- A sample space Omega containing all possible elementary outcomes.
- A collection of events, where each event is a set of outcomes.
- A probability function P that assigns a number P(A) to each event A.

Classical and quantum probability

Classical probability begins with probabilities assigned to outcomes or hidden alternatives. Quantum theory begins with a state vector or density matrix and uses the Born rule to generate probabilities. Once the probabilities are generated, the observed counts obey ordinary probability theory.

Quantum state / measurement choice --Born rule--> probability distribution --sampling--> observed counts

This distinction is essential: amplitudes can interfere, but the final experimental data are ordinary random outcomes.

2. Sample spaces and events

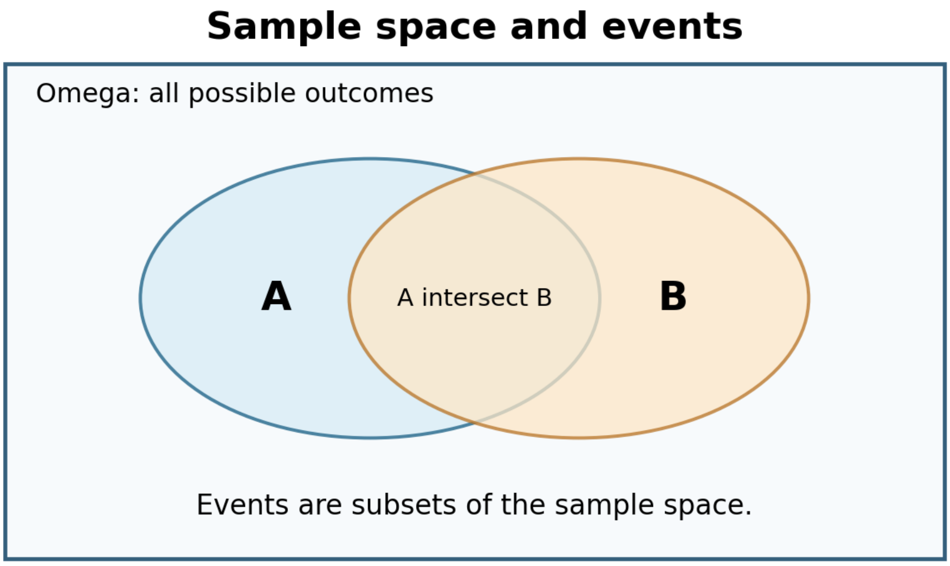


Figure 1. Events are subsets of a sample space.

For a single computational-basis measurement of one qubit, the sample space is

$$\Omega = \{0, 1\}.$$

For two measured qubits, the sample space is

$$\Omega = \{00, 01, 10, 11\}.$$

An event may contain one or more outcomes. For example, for two bits the event "the two outcomes are equal" is

$$E = \{00, 11\}.$$

Elementary and composite events

An elementary event contains one outcome, such as $\{01\}$. A composite event contains several outcomes, such as $\{00,11\}$. The impossible event is the empty set, and the certain event is the whole sample space.

Quantum experiment example

Prepare a qubit, apply a circuit, and measure it 1000 times. Each run produces one elementary outcome in $\{0,1\}$. The entire histogram is a sample from one probability distribution determined by the state and measurement basis.

3. Probability axioms and event algebra

A probability function satisfies the Kolmogorov axioms:

1. Non-negativity: P(A) >= 0 for every event A.
2. Normalization: P(Omega) = 1.
3. Additivity: if A and B cannot occur together, then P(A union B) = P(A) + P(B).

Useful consequences

P(empty set) = 0
P(not A) = 1 - P(A)
P(A union B) = P(A) + P(B) - P(A intersect B)

The subtraction prevents outcomes in the overlap from being counted twice.

Example

Suppose P(A)=0.6, P(B)=0.5, and P(A intersect B)=0.3. Then

P(A union B) = 0.6 + 0.5 - 0.3 = 0.8.

Mutually exclusive is not the same as independent

Mutually exclusive events have no shared outcomes. Independent events do not influence each other. Two nontrivial mutually exclusive events are usually not independent because observing one makes the other impossible.

4. Conditional probability and independence

Conditional probability updates a probability after learning that another event occurred.

$$P(A \mid B) = P(A \text{ intersect } B) / P(B), \text{ provided } P(B) > 0.$$

The condition B restricts the sample space to the outcomes in B.

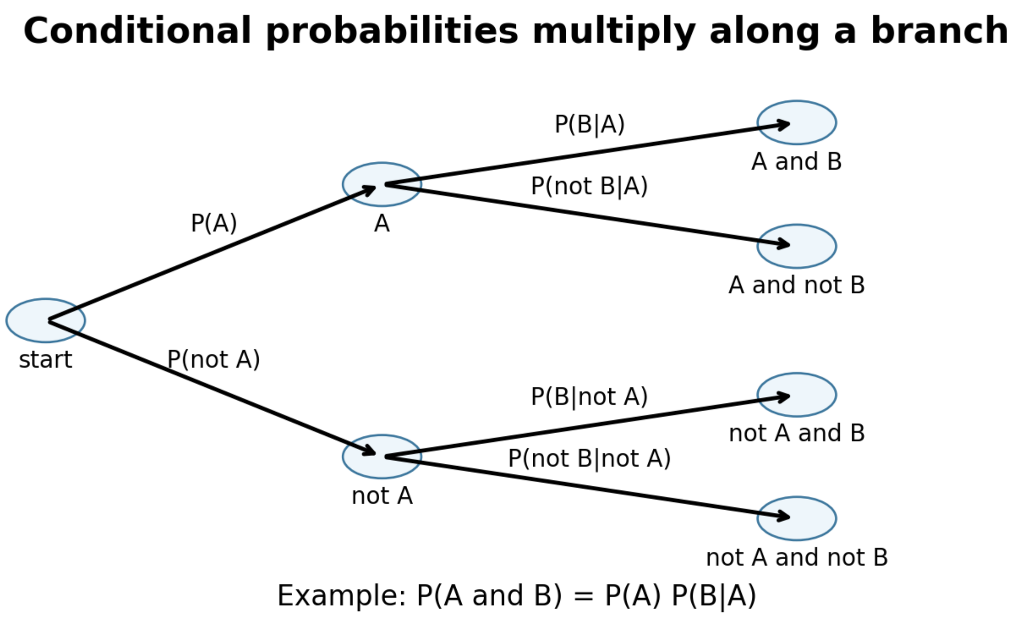


Figure 2. Probabilities multiply along a branch of a probability tree.

Multiplication rule

$$P(A \text{ intersect } B) = P(A)\, P(B \mid A) = P(B)\, P(A \mid B).$$

Independence

Events A and B are independent when learning one does not change the probability of the other:

$$P(A \mid B) = P(A).$$

Equivalently,

$$P(A \text{ intersect } B) = P(A)P(B).$$

Repeated circuit shots are often modeled as independent and identically distributed trials. Real hardware can violate this approximation through drift, memory effects, leakage, and correlated noise.

5. Bayes theorem

Bayes theorem reverses the direction of conditioning:

$$P(A \mid B) = P(B \mid A) P(A) / P(B).$$

When $\{A_k\}$ is a partition of the sample space, the denominator can be expanded using the law of total probability:

$$P(B) = \sum_k P(B \mid A_k) P(A_k).$$

Diagnostic example

Suppose a device is faulty with prior probability 0.02. A test reports positive with probability 0.95 when faulty and 0.04 when healthy.

$$P(\text{faulty} \mid \text{positive}) = 0.95(0.02) / [0.95(0.02) + 0.04(0.98)]$$

$$= 0.019 / 0.0582 \text{ approximately } 0.326.$$

Even a good test can produce a modest posterior probability when the condition is rare. This base-rate effect is important when interpreting calibration tests, error syndromes, and anomaly detectors.

Bayesian quantum applications

- Updating beliefs about an unknown state or parameter.
- Adaptive experiment design.
- Calibration and drift tracking.
- Decoding error syndromes using prior noise models.

6. Random variables and probability distributions

A random variable is a function that assigns a numerical value to each outcome. Despite the name, it is the outcome that is random; the function is fixed.

For a Z measurement, define

$$X(0)=+1,\ X(1)=-1.$$

Then X is a random variable taking values +1 and -1.

Probability mass function

For a discrete random variable X, the probability mass function is

$$p_X(x) = P(X=x).$$

It satisfies $p_X(x)>=0$ and $\sum_x p_X(x)=1$.

Cumulative distribution function

The cumulative distribution function is

$$F_X(x)=P(X<=x).$$

For quantum circuits measured as bit strings, the distribution is usually discrete. Continuous distributions become important for continuous-variable quantum systems, analog signals, timing, and parameter estimation.

Support

The support is the set of values with nonzero probability. A Pauli measurement has support {+1,-1}; a measured n-qubit register has support among 2^n bit strings.

7. Expectation, variance, and covariance

The expectation of a discrete random variable is its probability-weighted average:

$$E[X] = \sum_x x \, p_X(x).$$

It is the long-run average over many independent repetitions, not necessarily a value observed in one run.

Variance

$$\mathrm{Var}(X) = E[(X-E[X])^2] = E[X^2] - E[X]^2.$$

The standard deviation is $\sqrt{\mathrm{Var}(X)}$. It has the same units as X.

Covariance

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y].$$

Covariance measures linear statistical dependence. Independence implies zero covariance when the moments exist, but zero covariance does not generally imply independence.

Pauli-valued random variables

For a variable taking only +1 and -1, $X^2=1$ always. Therefore

$$\mathrm{Var}(X) = 1 - E[X]^2.$$

This formula is frequently used for shot-noise estimates of Pauli expectation values.

8. Bernoulli, binomial, and multinomial models

A Bernoulli trial has two outcomes: success with probability p and failure with probability 1-p. A single qubit measured as 0 or 1 is naturally modeled this way.

Binomial distribution

If N independent Bernoulli trials are performed and K counts the successes, then

$$P(K=k) = C(N,k)\, p^k (1-p)^{(N-k)}.$$

Its mean and variance are

$$E[K]=Np,\ Var(K)=Np(1-p).$$

Binomial distributions for repeated independent trials

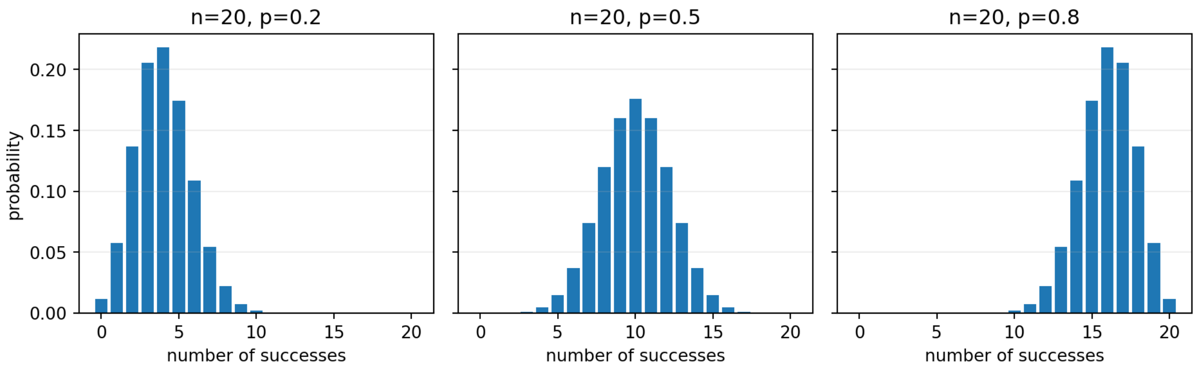


Figure 3. Shape of the binomial distribution for several probabilities.

Multinomial distribution

When each shot has m possible outcomes with probabilities p_1,...,p_m, the count vector follows a multinomial distribution. This is the natural model for n-qubit bit-string histograms.

$$P(N_1=n_1,\dots,N_m=n_m) = N! / (n_1!\dots n_m!)\ \mathrm{product}_j\ p_j^{(n_j)}.$$

9. Law of large numbers and finite shots

Let X_1,...,X_N be independent copies of a random variable with mean mu. Their sample mean is

X_bar_N = (1/N) sum_{k=1}^N X_k.

The law of large numbers states that X_bar_N approaches mu as N grows.

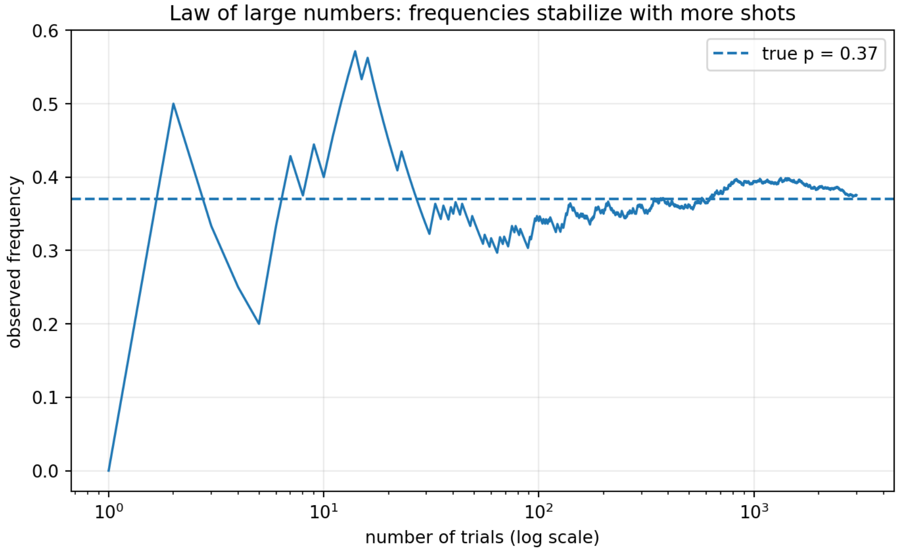


Figure 4. The observed frequency approaches the true probability, but it continues to fluctuate.

Quantum shots

A simulator may provide exact statevector probabilities. Hardware normally returns finite counts. If the true probability of 0 is p_0 and N shots are used, the count n_0 is random even when the device is perfectly stable.

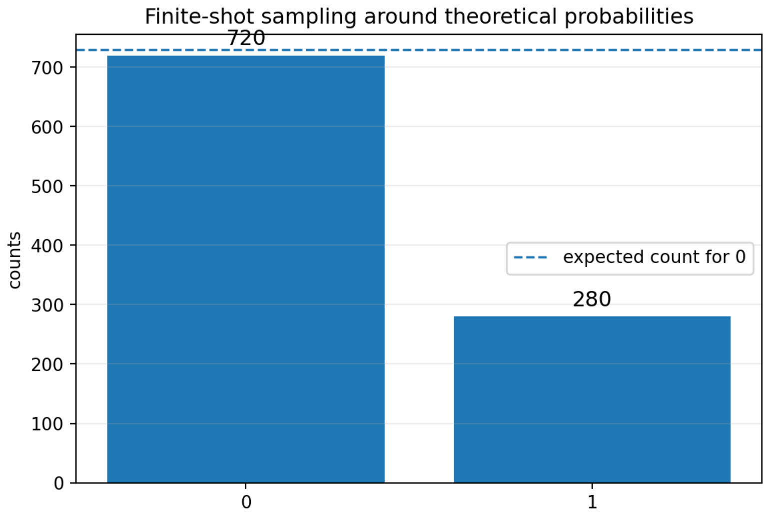


Figure 5. One finite sample from a distribution with p(0)=0.73.

10. Sampling error and confidence

The empirical frequency is

$$\hat{p} = K/N.$$

For a binomial model,

$$E[\hat{p}]=p,\ \mathrm{Var}(\hat{p})=p(1-p)/N.$$

The standard error is

$$SE(\hat{p})=\sqrt{p(1-p)/N}.$$

Since p is unknown, a simple estimate replaces p with p_hat. The worst case occurs near p=1/2, giving SE <= 1/(2 sqrt(N)).

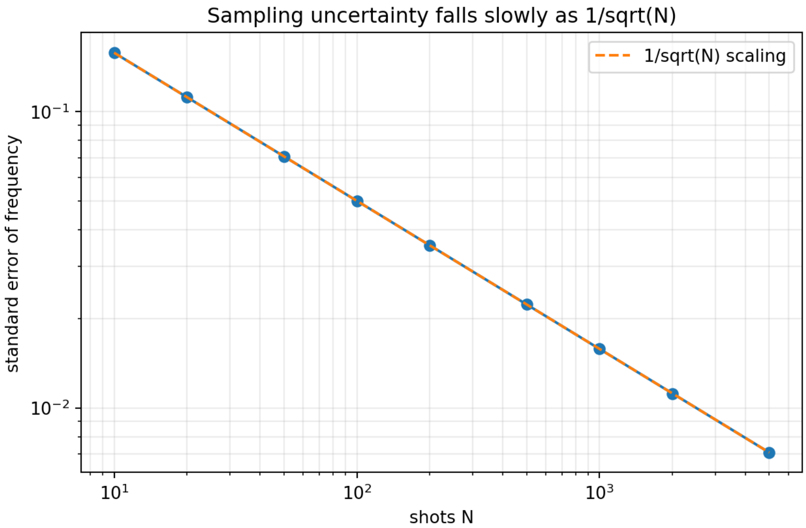


Figure 6. Reducing statistical error by a factor of 10 requires about 100 times as many shots.

Approximate confidence interval

For sufficiently large N and probabilities not too close to 0 or 1, a rough 95% interval is

$$\hat{p} \pm 1.96 \sqrt{\hat{p}(1-\hat{p})/N}.$$

Near boundaries or for small samples, Wilson or exact binomial intervals are preferable. Statistical uncertainty is separate from device bias, drift, gate error, and readout error.

11. Born rule

For a normalized state |psi> and a normalized measurement state |phi>, the Born rule gives

$$P(\phi) = |\langle\phi|\psi\rangle|^2.$$

The inner product is a complex amplitude. Its squared magnitude is a real nonnegative probability.

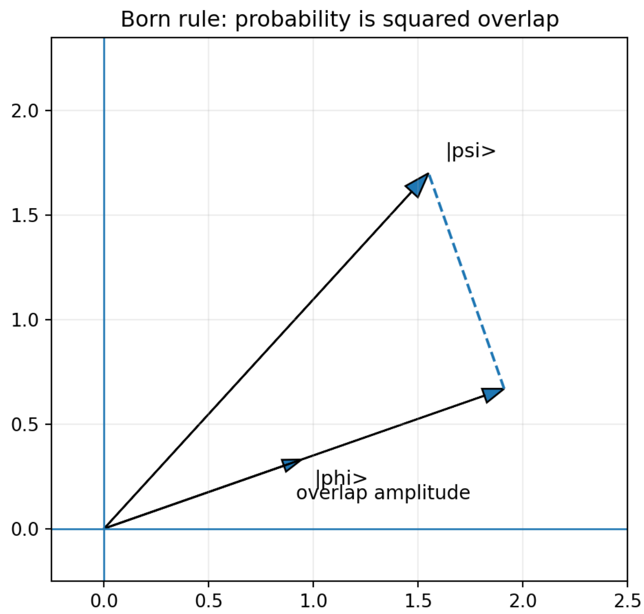


Figure 7. Quantum probability is the squared length of the projected component.

Computational basis

For

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

the amplitudes are

$$\langle 0|\psi\rangle = \alpha, \quad \langle 1|\psi\rangle = \beta,$$

so

$$P(0)=|\alpha|^2, \quad P(1)=|\beta|^2.$$

Normalization guarantees $P(0)+P(1)=1$.

12. Measurement in different bases

Probabilities depend on both the state and the measurement basis. For an orthonormal basis $\{|b_k\rangle\}$,

$$P(k)=|\langle b_k|\psi\rangle|^2.$$

Example: X basis

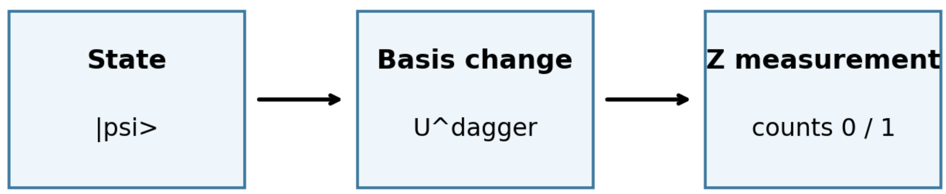
The X-basis states are

$$|+\rangle=(|0\rangle+|1\rangle)/\sqrt{2}, \quad |-\rangle=(|0\rangle-|1\rangle)/\sqrt{2}.$$

For $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$,

$$P(+)=|\alpha+\beta|^2/2, \quad P(-)=|\alpha-\beta|^2/2.$$

The sums and differences expose relative phase through interference.



To measure in another orthonormal basis, rotate that basis to the computational basis first.

Figure 8. A basis-change unitary followed by Z measurement implements measurement in another basis.

In circuits, X-basis measurement is implemented by H followed by computational-basis measurement. Y-basis measurement can be implemented by $S^\dagger$ then H, followed by computational-basis measurement.

13. Projective measurement and state update

A projective measurement is described by orthogonal projectors $\{P_k\}$ satisfying

$$P_k^2=P_k,\ P_k^\dagger=P_k,\ P_j P_k=0 \text{ for } j\neq k,\ \sum_k P_k=I.$$

Outcome probabilities are

$$p_k = \langle\psi|P_k|\psi\rangle.$$

Conditioned on obtaining outcome k, the post-measurement state is

$$|\psi_k\rangle = P_k|\psi\rangle / \sqrt{p_k}, \text{ when } p_k>0.$$

Example

For computational-basis measurement, $P_0=|0\rangle\langle 0|$ and $P_1=|1\rangle\langle 1|$. If $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$, outcome 0 occurs with probability $|\alpha|^2$ and the conditioned state becomes $|0\rangle$.

Unconditioned measurement

If the outcome is ignored, a projective measurement generally produces a mixed state:

$$\rho \to \sum_k P_k \rho P_k.$$

This removes coherence between different measurement subspaces.

14. Expectation values of quantum observables

For a Hermitian observable A and state |psi>, the expected measurement result is

$$\langle A \rangle = \langle \psi | A | \psi \rangle.$$

If A has spectral decomposition A=sum_k a_k P_k, then

$$\langle A \rangle = \sum_k a_k p_k.$$

This is exactly the classical expectation of the random measurement outcome.

Pauli Z example

For |psi>=alpha|0>+beta|1> and Z eigenvalues +1,-1,

$$\langle Z \rangle = |\alpha|^2 - |\beta|^2.$$

From counts n_0,n_1 over N shots, estimate

$$\widehat{\langle Z \rangle} = (n_0 - n_1)/N.$$

Variance

$$\mathrm{Var}(A) = \langle A^2 \rangle - \langle A \rangle^2.$$

For Pauli observables A^2=I, so Var(A)=1-<A>^2. An eigenstate has zero variance because every repetition returns the same eigenvalue.

15. Joint, marginal, and conditional distributions

For two measured systems A and B, the joint distribution assigns probabilities P(a,b).

Marginal probabilities are obtained by summing over the other variable:

$$P_A(a)=\sum_b P(a,b),\ P_B(b)=\sum_a P(a,b).$$

Conditional probabilities are

$$P(a|b)=P(a,b)/P_B(b).$$

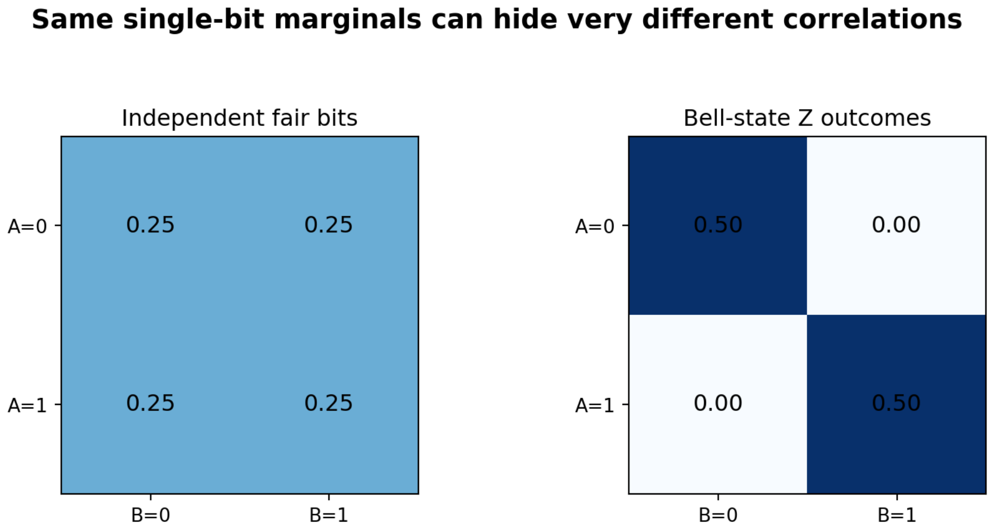


Figure 9. Uniform independent bits and Bell-state outcomes share the same single-bit marginals but not the same joint distribution.

For the Bell state

$$|\Phi^+\rangle = (|00\rangle+|11\rangle)/\sqrt{2},$$

the Z-basis joint probabilities are P(00)=P(11)=1/2 and P(01)=P(10)=0. Each qubit alone is uniformly random, yet their outcomes are perfectly correlated.

16. Correlations and entanglement

Correlation measures how joint outcomes differ from what would be expected from independent marginals.

For numerical variables X and Y,

$$\mathrm{Cov}(X,Y)=E[XY]-E[X]E[Y].$$

For Pauli measurements on two qubits, a common correlation is

$$\langle Z \text{ tensor } Z\rangle = \text{sum}_{a,b \text{ in } \{+1,-1\}} ab\, P(a,b).$$

Bell-state example

For $|\mathrm{Phi}+\rangle$, the Z outcomes are always equal. Therefore Z tensor Z always has value +1, so

$$\langle Z \text{ tensor } Z\rangle=1.$$

Yet

$$\langle Z \text{ tensor } I\rangle=0,\ \langle I \text{ tensor } Z\rangle=0.$$

Each subsystem is individually unbiased, while the pair is strongly correlated.

Correlation is not automatically entanglement

Classical mixtures can also be correlated. Entanglement is a stronger structural property: the joint quantum state cannot be written as a probabilistic mixture of product states. Demonstrating it generally requires measurements in more than one basis or an entanglement witness.

17. Classical mixtures versus coherent superpositions

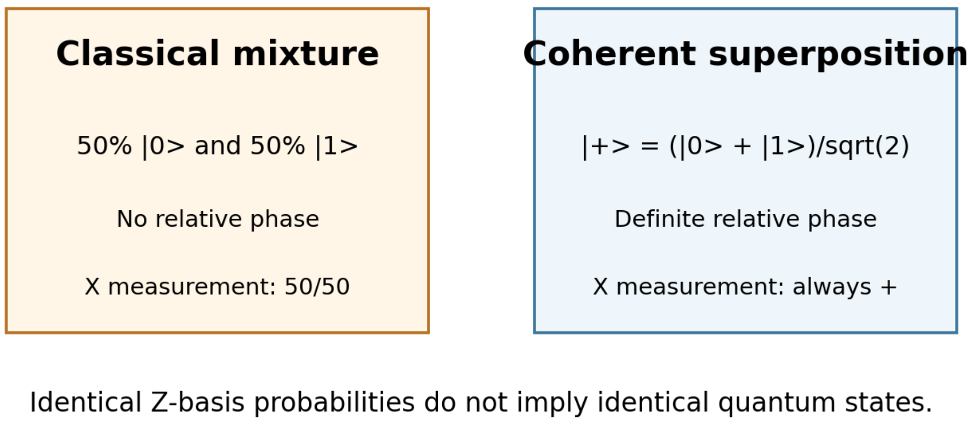


Figure 10. Equal Z-basis probabilities can arise from physically different states.

The coherent state

|+>=(|0>+|1>)/sqrt(2)

and the classical mixture "50% |0>, 50% |1>" both produce 0 and 1 with equal probability in the Z basis.

However, in the X basis:

- |+> produces the + outcome with probability 1.
- The classical mixture produces + and - with probability 1/2 each.

The difference is coherence: the superposition contains a definite relative phase, while the mixture represents classical uncertainty over alternatives.

This is why a list of probabilities in one basis is not a complete description of a quantum state.

18. Density-matrix probability rules

A density matrix rho describes pure states, mixed states, and reduced states of larger systems. It is Hermitian, positive semidefinite, and has trace 1.

For a measurement operator represented by projector P_k,

$$p_k = Tr(rho\ P_k).$$

For a pure state $rho=|psi><psi|$, this reduces to

$$Tr(|psi><psi|P_k) = <psi|P_k|psi>.$$

Expectation values become

$$<A> = Tr(rho\ A).$$

Example: incoherent qubit mixture

$$rho_{mix} = (1/2)|0><0| + (1/2)|1><1| = I/2.$$

For any Pauli observable X,Y,Z,

$$Tr(rho_{mix}\ X)=Tr(rho_{mix}\ Y)=Tr(rho_{mix}\ Z)=0.$$

This density matrix represents complete uncertainty about every qubit direction, unlike a pure state that may be random in one basis but deterministic in another.

19. Readout noise and confusion matrices

Measured counts may differ from the ideal distribution because the detector can report the wrong classical outcome. A simple readout model uses a stochastic matrix M:

$$p\_observed = M\ p\_true.$$

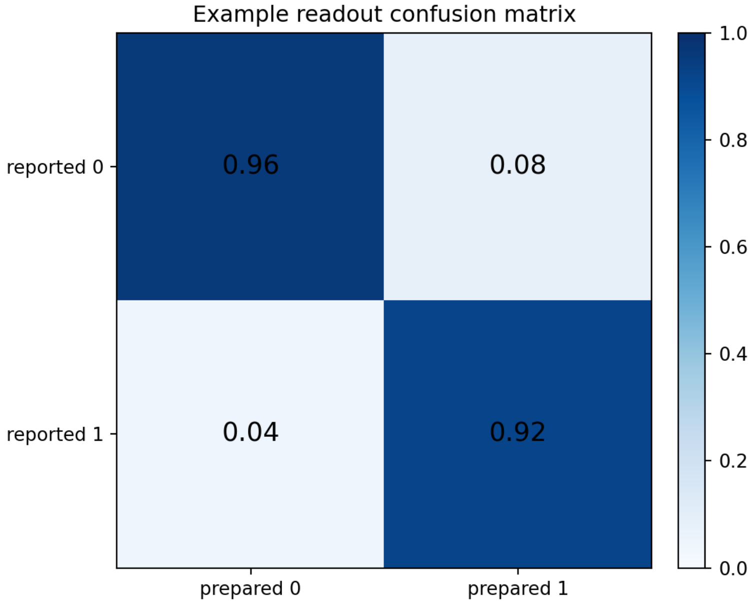


Figure 11. Columns represent prepared states and rows represent reported outcomes.

In the example, a prepared 0 is reported as 0 with probability 0.96 and as 1 with probability 0.04. A prepared 1 is reported correctly with probability 0.92.

Mitigation idea

Estimate M using calibration circuits, then solve approximately for p_true. Direct inversion can amplify noise and may produce negative estimates, so practical methods use constrained estimation or regularization.

Beyond readout error

Gate noise, decoherence, leakage, crosstalk, drift, and state-preparation error change the underlying distribution before measurement. More shots reduce sampling noise but do not remove systematic bias.

20. Statistical estimation in quantum experiments

A quantum experiment usually estimates parameters from repeated outcomes. An estimator is a function of the observed data.

Estimating a probability

$$\hat{p}=n/N.$$

Estimating a Pauli expectation

$$\hat{\mu}=(n_+ - n_-)/N = 2(n_+/N)-1.$$

For true expectation mu, the variance of this estimator under independent shots is

$$\mathrm{Var}(\hat{\mu}) = (1-\mu^2)/N.$$

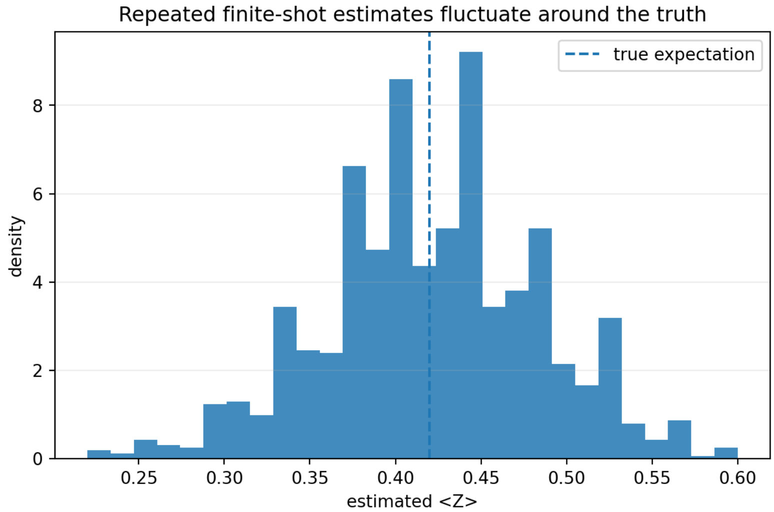


Figure 12. Repeating an N-shot experiment gives different estimates around the true value.

Bias and consistency

- An estimator is unbiased if its expected value equals the parameter.
- It is consistent if it converges to the true value as the sample size grows.
- Low variance is useful, but a low-variance biased estimator can still be misleading.

Shot allocation

Variational algorithms often estimate many Pauli terms. Efficiently allocating shots among terms can reduce total uncertainty. Terms with larger coefficients or larger intrinsic variance generally deserve more shots.

21. Worked examples

Example 1: computational-basis probabilities

For

$$|\psi\rangle = (\sqrt{3}/2)|0\rangle + (i/2)|1\rangle,$$

the probabilities are

$$P(0)=3/4,\ P(1)=1/4.$$

The phase i does not change $P(1)$, but it can affect measurements in another basis.

Example 2: X-basis measurement

For $|\psi\rangle=(|0\rangle+i|1\rangle)/\sqrt{2}$,

$$\langle +|\psi\rangle = (1+i)/2,\ \langle -|\psi\rangle = (1-i)/2.$$

Both squared magnitudes are 1/2, so $P(+)=P(-)=1/2$.

Example 3: finite-shot estimate

Suppose a Z measurement gives $n_0=730$ and $n_1=270$ from $N=1000$ shots.

$$\hat{p}(0)=0.730,\ \widehat{\langle Z\rangle}=(730-270)/1000=0.460.$$

The estimated standard error of $\hat{p}$ is approximately

$$\sqrt{0.73(0.27)/1000} \text{ approximately } 0.014.$$

Worked examples continued

Example 4: Bell-state correlation

For |Phi+>=(|00>+|11>)/sqrt(2), the Z-basis probabilities are 1/2 for 00 and 1/2 for 11. Therefore each qubit has marginal distribution (1/2,1/2), but the parity is always even.

<Z tensor Z> = (+1)(1/2) + (+1)(1/2) = 1.

Example 5: Bayes update for readout

Suppose P(prepared 1)=0.3, P(report 1|prepared 1)=0.92, and P(report 1|prepared 0)=0.04.

P(prepared 1 | report 1) = 0.92(0.3) / [0.92(0.3)+0.04(0.7)]

= 0.276 / 0.304 approximately 0.908.

Example 6: expectation from spectral probabilities

An observable has eigenvalues -2, 1, and 4 with probabilities 0.2, 0.5, and 0.3.

E[A] = (-2)(0.2)+(1)(0.5)+(4)(0.3)=1.3.

E[A^2] = 4(0.2)+1(0.5)+16(0.3)=6.1.

Var(A)=6.1-(1.3)^2=4.41.

22. Common mistakes

Mistake	Correction
Treating amplitude as probability	Probability is the squared magnitude of an amplitude.
Adding probabilities before amplitudes interfere	For indistinguishable coherent alternatives, add amplitudes first and then square.
Assuming more shots remove all error	More shots reduce sampling variance, not systematic device bias.
Confusing independent and mutually exclusive events	Independent events may occur together; mutually exclusive events cannot.
Using P(A|B)=P(B|A)	Conditional probabilities are generally different; Bayes theorem connects them.
Interpreting an expectation value as a single shot outcome	An expectation is a repeated-trial average.
Assuming identical marginals imply identical joint states	Different correlations can share the same marginals.
Equating a 50/50 mixture with |+>	They match in the Z basis but differ in the X basis.
Ignoring basis choice	A state can be deterministic in one basis and random in another.
Reporting only shot noise	Calibration uncertainty, drift, and model error may dominate.

A useful habit

Whenever you see a probability statement, identify: the sample space, the event or random variable, the conditioning information, the assumed independence structure, and whether the number is theoretical, estimated from data, or corrected by a noise model.

23. Exercises

Foundations

1. A two-qubit measurement has sample space {00,01,10,11}. Let A be the event that the first bit is 1 and B the event that the parity is even. List A, B, A intersect B, and A union B.
2. Given P(A)=0.55, P(B)=0.40, and P(A intersect B)=0.20, calculate P(A union B), P(A|B), and P(B|A). Are A and B independent?
3. A disease has prior probability 0.01. A test has sensitivity 0.98 and false-positive rate 0.03. Find P(disease|positive).

Random variables

1. A variable X takes values -1 and +1 with P(+1)=0.7. Find E[X], Var(X), and the standard deviation.
2. For N=500 shots and true probability p=0.4, find the expected success count, count variance, and standard error of the observed frequency.
3. A three-outcome measurement has probabilities (0.2,0.5,0.3). In 1000 shots, what count vector is expected? Are the actual counts guaranteed to equal it?

Quantum probability

1. For |psi>=(|0>+sqrt(3)i|1>)/2, find Z-basis probabilities.
2. For the same state, find X-basis probabilities.
3. For |psi>=cos(theta/2)|0>+e^(i phi) sin(theta/2)|1>, show that <Z>=cos(theta).
4. For |Phi+>, find the probabilities of the four Z-basis outcomes and the marginal distribution of each qubit.

Exercises continued

Measurement and estimation

1. A Z experiment returns counts {0: 615, 1: 385}. Estimate P(0), P(1), and <Z>.
2. Estimate the standard error of P(0) in the previous exercise.
3. An observable has eigenvalues 0,2,5 with probabilities 0.1,0.6,0.3. Find its expectation and variance.
4. A projector P has probability p=<psi|P|psi>=0.16. What is the norm of P|psi>?
5. Explain why measuring |+> in the Z basis gives a random result while measuring it in the X basis is deterministic.
6. Explain why a Bell state can have uniform single-qubit marginals and still contain strong correlations.
7. A readout matrix is M=[[0.95,0.10],[0.05,0.90]]. If the true distribution is (0.8,0.2)^T, calculate the observed distribution.
8. Describe the difference between sampling uncertainty, readout bias, and gate noise.

Challenge problems

1. Derive Var(mu_hat)=(1-mu^2)/N for a Pauli-valued measurement estimator.
2. Show that if X and Y are independent, then E[XY]=E[X]E[Y]. State why the converse is not always true.
3. For a normalized state |psi> and complete orthonormal basis {|b_k>}, prove sum_k |<b_k|psi>|^2=1.
4. Compare the X-basis measurement probabilities of rho=I/2 and rho=|+><+|.

24. Selected solutions

Exercise 1

A={10,11}; B={00,11}; A intersect B={11}; A union B={00,10,11}.

Exercise 2

P(A union B)=0.55+0.40-0.20=0.75.

P(A|B)=0.20/0.40=0.50, P(B|A)=0.20/0.55 approximately 0.364.

Since P(A)P(B)=0.22 is not 0.20, the events are not independent.

Exercise 4

E[X]=(+1)(0.7)+(-1)(0.3)=0.4.

Var(X)=1-(0.4)^2=0.84, sigma=sqrt(0.84) approximately 0.917.

Exercise 7

The amplitudes are 1/2 and sqrt(3)i/2, so

P(0)=1/4, P(1)=3/4.

Exercise 11

P_hat(0)=0.615, P_hat(1)=0.385, <Z>_hat=0.230.

Selected solutions continued

Exercise 12

$$SE \approx \sqrt{0.615(0.385)/1000} \approx 0.0154.$$

Exercise 13

$$E[A]=0(0.1)+2(0.6)+5(0.3)=2.7.$$

$$E[A^2]=0+4(0.6)+25(0.3)=9.9.$$

$$Var(A)=9.9-(2.7)^2=2.61.$$

Exercise 14

Because $p=\langle\psi|P|\psi\rangle=||P|\psi\rangle||^2$, the projected norm is $\sqrt{0.16}=0.4$.

Exercise 17

$$p_{observed} = [[0.95,0.10],[0.05,0.90]]\,[0.8,0.2]^T = [0.78,0.22]^T.$$

Challenge 1

Let each Pauli outcome X_i be ± 1 with mean μ. Then $Var(X_i)=1-\mu^2$. The sample mean $\hat{\mu}=(1/N)\sum_i X_i$ has variance

$$Var(\hat{\mu})=(1/N^2) \sum_i Var(X_i)=(1-\mu^2)/N$$

when shots are independent.

25. Compact reference sheet

<table>
<tr><th>Concept</th><th>Formula</th></tr>
<tr><td>Complement</td><td>P(not A)=1-P(A)</td></tr>
<tr><td>Union</td><td>P(A union B)=P(A)+P(B)-P(A intersect B)</td></tr>
<tr><td>Conditional probability</td><td>P(A|B)=P(A intersect B)/P(B)</td></tr>
<tr><td>Independence</td><td>P(A intersect B)=P(A)P(B)</td></tr>
<tr><td>Bayes theorem</td><td>P(A|B)=P(B|A)P(A)/P(B)</td></tr>
<tr><td>Expectation</td><td>E[X]=sum_x x p(x)</td></tr>
<tr><td>Variance</td><td>Var(X)=E[X^2]-E[X]^2</td></tr>
<tr><td>Covariance</td><td>Cov(X,Y)=E[XY]-E[X]E[Y]</td></tr>
<tr><td>Binomial probability</td><td>P(K=k)=C(N,k)p^k(1-p)^(N-k)</td></tr>
<tr><td>Frequency standard error</td><td>sqrt[p(1-p)/N]</td></tr>
<tr><td>Born rule</td><td>P(phi)=|<phi|psi>|^2</td></tr>
<tr><td>Projective probability</td><td>p_k=<psi|P_k|psi></td></tr>
<tr><td>Post-measurement state</td><td>P_k|psi>/sqrt(p_k)</td></tr>
<tr><td>Observable expectation</td><td><A>=<psi|A|psi>=Tr(rho A)</td></tr>
<tr><td>Observable variance</td><td><A^2>-<A>^2</td></tr>
<tr><td>Pauli estimate</td><td>mu_hat=(n_+-n_-)/N</td></tr>
<tr><td>Joint marginal</td><td>P_A(a)=sum_b P(a,b)</td></tr>
<tr><td>Density-matrix Born rule</td><td>p_k=Tr(rho P_k)</td></tr>
</table>

Final perspective

Probability enters quantum computing at two distinct levels. First, complex amplitudes and measurement operators generate an exact theoretical probability distribution through the Born rule. Second, finite experiments sample from that distribution, introducing ordinary statistical uncertainty. Keeping these levels separate prevents many conceptual and practical errors.

The next topic in the roadmap is classical bits and information foundations, followed by qubits and quantum measurement.